

```
In [9]: %config SqlMagic.style = 'DEFAULT'
```

EDA

1- Sales by Country

2- Country Segmentation

```
In [10]: %%sql
select top(5) * from dbo.adding

* mssql+pyodbc://@AAAA\MSSQLSERVER02/Amira?driver=ODBC+Driver+17+for+SQL+Server&tr
sted_connection=yes
Done.
```

```
Out[10]: InvoiceNo  StockCode  Description  Quantity  InvoiceDate  UnitPrice  CustomerID  Country
```

536365	85123A	WHITE HANGING HEART T- LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850	Unitec Kingdom
536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850	Unitec Kingdom
536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850	Unitec Kingdom
536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850	Unitec Kingdom
536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	17850	Unitec Kingdom

Creating a Country Segmentation View

```
In [11]: %%sql

create view Country_Level as
select country, round(sum( Quantity* UnitPrice), 0) total_sales,
```

```

case
    when sum( Quantity* UnitPrice)> 100000 then 'vip country'
    when sum( Quantity* UnitPrice)> 50000 then 'average country'
    else 'Normal country'
end as Country_Levels

from dbo.adding
group by country
order by total_sales desc

```

```

* mssql+pyodbc://@AAAA\MSSQLSERVER02/Amira?driver=ODBC+Driver+17+for+SQL+Server&trusted_connection=yes
(pyodbc.ProgrammingError) ('42000', '[42000] [Microsoft][ODBC Driver 17 for SQL Server][SQL Server]The ORDER BY clause is invalid in views, inline functions, derived tables, subqueries, and common table expressions, unless TOP, OFFSET or FOR XML is also specified. (1033) (SQLExecDirectW)')
[SQL: create view Country_Level as
select country, round(sum( Quantity* UnitPrice), 0) total_sales,
case
    when sum( Quantity* UnitPrice)> 100000 then 'vip country'
    when sum( Quantity* UnitPrice)> 50000 then 'average country'
    else 'Normal country'
end as Country_Levels

from dbo.adding
group by country
order by total_sales desc]
(Background on this error at: https://sqlalche.me/e/20/f405)

```

Sales by Country

In [12]: %%sql

```
select top(5)* from Country_level
```

```

* mssql+pyodbc://@AAAA\MSSQLSERVER02/Amira?driver=ODBC+Driver+17+for+SQL+Server&trusted_connection=yes
Done.

```

Out[12]:

country	total_sales	Country_Levels
Australia	137077.0	vip country
Austria	10154.0	Normal country
Bahrain	548.0	Normal country
Belgium	40911.0	Normal country
Brazil	1144.0	Normal country

Country Segmentation

In [13]: %%sql

```
select Country_Levels, count(country) as country_count
```

```
from Country_Level  
group by Country_Levels  
order by country_count desc
```

```
* mssql+pyodbc://@AAAA\MSSQLSERVER02/Amira?driver=ODBC+Driver+17+for+SQL+Server&tru  
sted_connection=yes  
Done.
```

Out[13]:

Country_Levels	country_count
Normal country	30
vip country	6
average country	2

In []: